IBM Z DATATHON 2026 · REAL-TIME AI FOR CRITICAL DECISIONS

Kavach

Stop the UPI scam **before the money moves**.

Real-time, explainable, privacy-preserving scam detection that runs inside the payment, on IBM LinuxONE.

Team: _____ · Bengaluru hub

AI for Good

THE PROBLEM

UPI made payments instant. It made scams instant too.

"Digital arrest"

Fake police / CBI keep a victim, often elderly, on a call until they transfer their savings.

Collect-request scam

"Approve to receive your ₹4,999 prize." Approving actually **sends** money.

SIM-swap takeover

New phone, other city: the account is drained in minutes.

Money mules

Rented accounts collect from many victims and pass the money on within minutes.

10+ billion UPI payments a month. These scams beat people, not encryption.

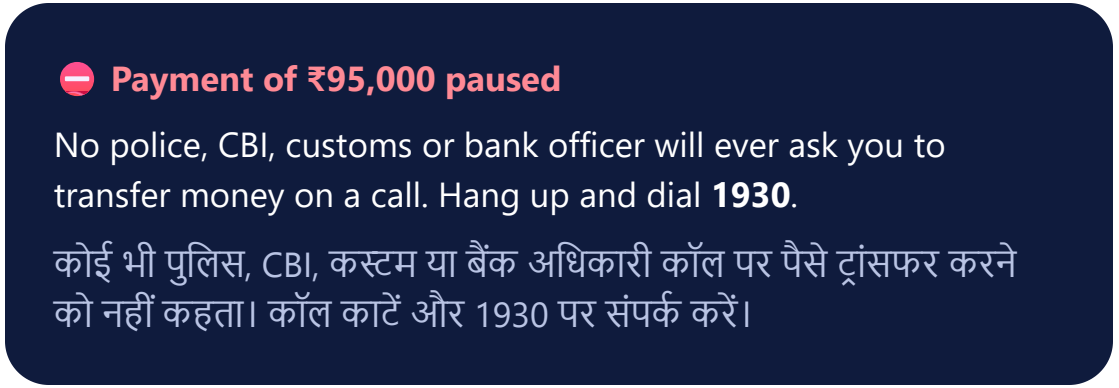
Once a payment settles, recovery is slow and often impossible. **The only moment to stop it is during the transaction.**

THE SOLUTION

Every payment gets a decision, and a reason, in ~13 ms

ALLOW	Goes through untouched: **99.7%** of genuine payments
STEP-UP	Clear warning + re-confirm. Medium risk or a never-seen pattern
HOLD	Payment paused, customer told why, can be released after verification

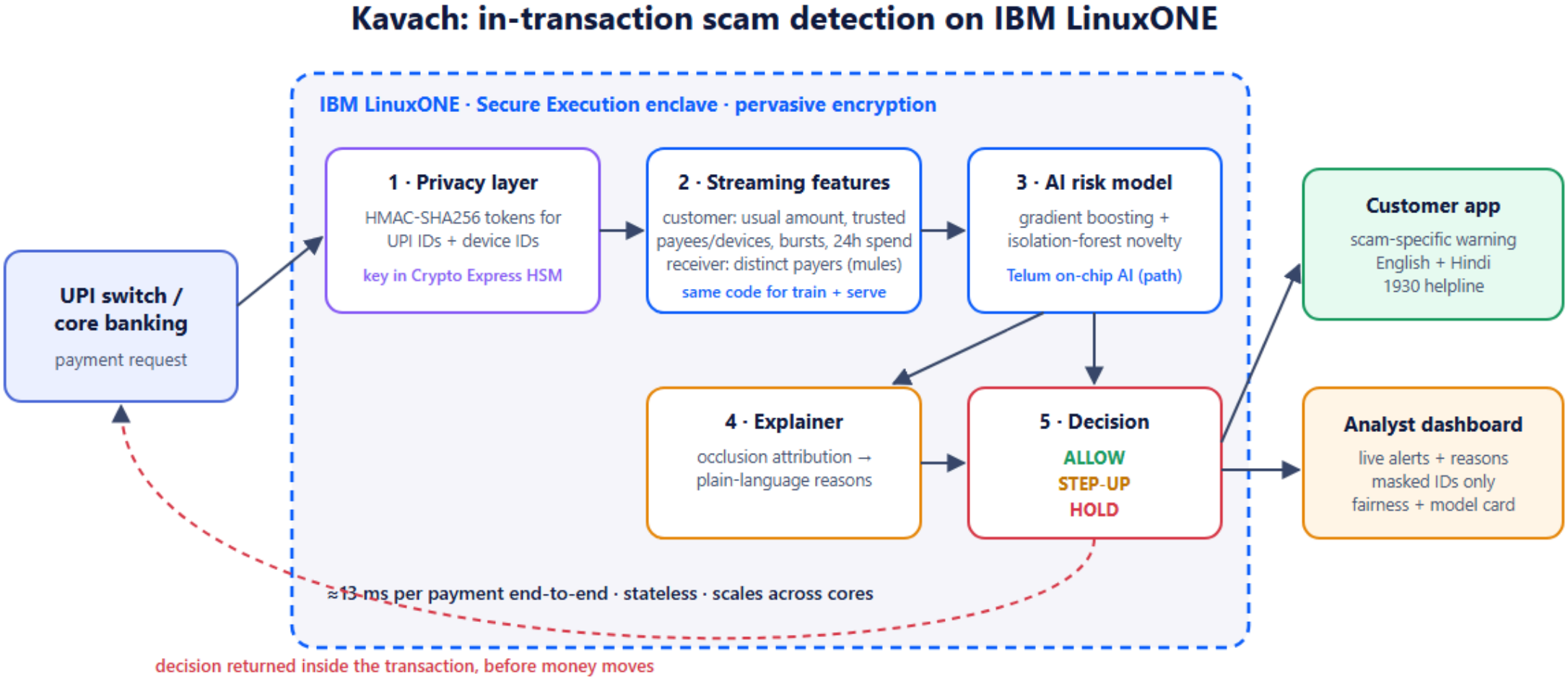
Every alert says **why**: "*on a live call while paying*", "*receiver got money from 7 people today*", "*device never seen on this account*".



What the customer sees: scam-specific, in English + Hindi, at the moment it matters.

ARCHITECTURE

AI where the transaction lives



HOW THE AI WORKS

Behaviour, not blacklists

Streaming feature store

14 live signals per payment: amount vs usual, trusted payees / devices / cities (trust takes 24 h), bursts, 24 h spend, on-call, COLLECT, receiver fan-in. **Same code trains and serves**, so there is no skew.

Two models

Gradient-boosted trees learn known scam patterns. An Isolation Forest trained only on genuine traffic flags **never-seen-before** behaviour for a STEP-UP.

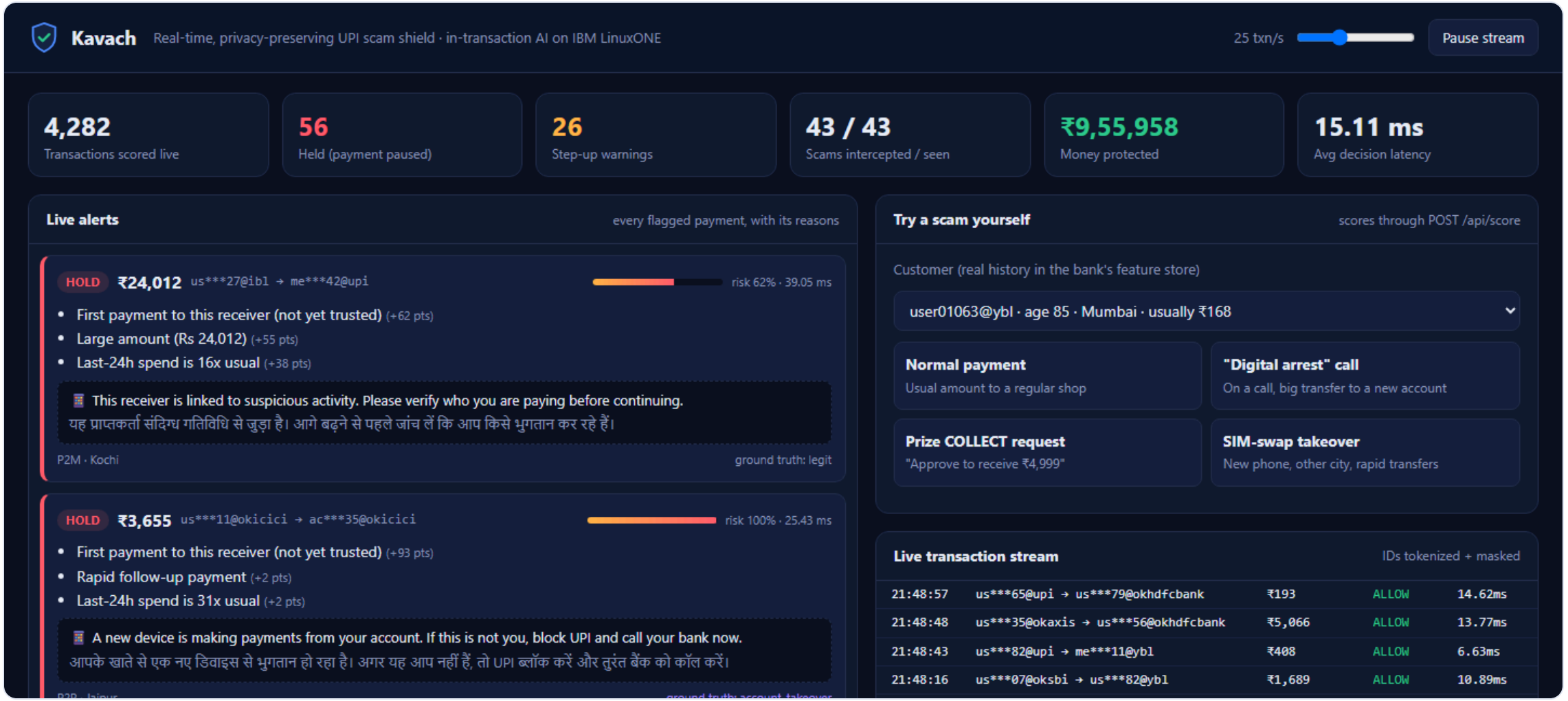
Explainer + policy

Occlusion attribution (one batched call) ranks the signals behind each alert. Thresholds tuned for ≥90% HOLD precision and ~97% warning recall.

Data: 140k simulated UPI payments (4,000 customers, 600 merchants, 80 mules, 30 days) with 4 fraud patterns from public NPCI / RBI / I4C advisories, plus hard negatives (rent to new payees, calls, travel, new shops, old rented mule accounts). **Time-split**: train on weeks 1–2, tune on days 18–22, test on the final unseen week.

LIVE DEMO

Analyst dashboard: live stream, alerts with reasons, try-a-scam



RESULTS · HELD-OUT WEEK THE MODEL NEVER SAW

Catches almost every scam, barely touches honest users

95.4%
of scams warned or held

98.1%
of fraud money intercepted (₹91.1 L of ₹92.9 L)

0.34%
of genuine payments see any friction (0.06% held)

13 ms
p50 end-to-end decision (p99 27 ms)

Scam type	Caught
Account takeover	97.7%
Money-mule cash-out	98.0%
Digital arrest	92.9%
Collect-request scam	92.9%

Model quality	
HOLD precision	95.2%
PR-AUC	0.972
ROC-AUC	0.999
Test set	32,854 txns · 437 scams

RESPONSIBLE AI · SECURITY BY DESIGN

Useful AI without exposing anyone

- **Identity never exposed:** UPI and device IDs are HMAC-tokenized at ingestion. The model, features and logs never see them. Analysts see masks only.
- **No demographics in the model.** Age is used only to audit outcomes.
- **Explainable by default:** no alert without its reasons.
- **Human in the loop:** HOLD pauses the payment, it doesn't cancel it.

Age group	Genuine flagged	Scams caught
18–35	0.28%	92.7%
36–55	0.38%	96.7%
56+	0.34%	95.7%

Older customers, the main targets, are protected as well as everyone else and are not over-blocked.

WHY IBM Z / LINUXONE

Bring the AI to the data, not the data to the AI

Need	IBM Z capability	Kavach
Score inside the payment, at UPI scale	Telum on-chip AI accelerator · Snap ML / zDLC · Machine Learning for z/OS (CICS/IMS)	Runs on LinuxONE today; single-row in-transaction path; tree model → accelerator next
Keep identities secret	Crypto Express HSM · pervasive encryption	HMAC tokenization at ingestion; key moves into the HSM
Protect model + data from insiders	IBM Secure Execution · Hyper Protect containers	Stateless container, ready for an enclave
Future-proof crypto	Quantum-safe ML-KEM / ML-DSA	Quantum-safe TLS + key wrapping, no model change
Never go down	Z reliability, scale-up per LPAR	Stateless scoring, scales across cores

Banks' transactions **already run on IBM Z**. Scoring them there removes a network hop, a copy of sensitive data and a point of failure.

IMPACT & NEXT STEPS

From prototype to every UPI payment

Who benefits

Elderly and first-time digital users, banks & PSPs (fewer losses, refunds and complaints), and law enforcement (1930 / I4C gets mule signals early).

Next 3 months

Model on the Telum accelerator (Snap ML / zDLC) · keys in Crypto Express · Secure Execution enclave · pilot on anonymised bank data.

Then

Shared mule-account signals across banks (privacy-preserving) · more Indian languages · drift + fairness monitoring.

Kavach: stop the scam before the money moves.

github.com/hsbofficial1/kavach